

Lab 03 - Structured Prompting for Cinematic Stills

Course: Create Short Video Film using AI (TGS-2020505925) | Version v12.0 | 27 September 2026 Topic 02: AI Image Processing, Generation and Visual Enhancement Outcome: ELO2 | In-class time: 15 minutes of scheduled practical time

Competency	Statement
Ability A2	Apply the principles of processing, filtering and analysis methods for video data
Knowledge K3	Methods to represent image and video data
Knowledge K4	Image and video processing, filtering and transformation methods

Scenario

Shot S2 - MEI runs through the night market - keeps coming back different: the coat changes colour, she jumps around the frame, the look shifts. You lint a weak six-field prompt, rewrite it, and measure four prompt designs (A: free prose → D: subject, look, camera and constraints locked) on four takes each.

What you will produce

A linted v1 prompt with its problems listed, your corrected v2 prompt in prose and JSON, and a variance table showing how much each prompt design reduced wardrobe, blocking, framing and look drift.

Tools: Python 3 | opencv-python | NumPy | optional: any image generator for your own takes

Environment

Install these once, before class if you can - the lab itself needs no network access after the dependencies are present:

```
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\Activate.ps1
pip install --upgrade pip
pip install "opencv-python>=4.10" numpy
python3 -c "import cv2, numpy; print(cv2.__version__, numpy.__version__)"
```

Required: opencv-python>=4.10, numpy.

API currency. Everything taught here runs on the base opencv-python wheel plus NumPy - no contrib modules and no downloaded model weights. SSIM is implemented in NumPy because cv2.quality is contrib-only. Generative steps are optional: every lab supplies SIMULATED reference media so it can be completed offline, and asks you to record the model ID and date if you generate your own.

Workflow

1. Lint the v1 prompt
2. Rewrite it as a six-field v2
3. Render it as prose, JSON and negative prompt

4. Measure variance across four designs
5. Record which field locks mattered most

Files in this folder

Path	What it is
reference/takes/cellA_01.png ... cellD_04.png	16 takes of shot S2: four prompt designs × four takes (SIMULATED, designed spread)
data/shot2-fields-v1.json	A weak six-field prompt with deliberate problems
data/shot2-fields-v2-template.json	The v2 template with the subject and references filled in
data/prompt-factors.json	What each design A-D locks, and what each measure means
prompt_lab.py	Runnable script - see the steps below
PROMPTS.md / PROMPTS.pdf	The reusable prompt and specification pack
out/	Everything you produce (created on first run)

Provenance. Every image and clip in reference/ is either a SIMULATED classroom illustration of *The Last Ferry*, rendered deterministically with OpenCV and labelled on the frame, or an AI-generated reference copied unchanged with a PROVENANCE.md beside it. Nothing in this folder may be relabelled: a simulated stand-in is never presented as model output or as a photograph.

Steps

1. Read data/prompt-factors.json: the six fields are subject, action, environment, camera, style and constraints. Each design A→D locks more of them.
2. Lint the v1 prompt. It exits with status 1 and lists every problem: a missing constraints field, 'night' vs 'noon', 'close-up' vs 'wide shot', a repeated word and no references.

```
python3 prompt_lab.py compose --fields data/shot2-fields-v1.json --out out
```

3. Copy data/shot2-fields-v2-template.json to out/shot2-fields-v2.json. Fill action, environment, camera, style and constraints. Use one camera description only (for example: medium shot, eye level, 35mm, tracking left to right).
4. Put the constraints in plain words: 'keep coat and scarf colours exactly as the reference sheet; no extra people in the foreground; subject stays in the centre third'.
5. Lint your v2 until it exits cleanly (status 0). Read the prose and JSON versions it writes and check the @element references come first.

```
python3 prompt_lab.py compose --fields out/shot2-fields-v2.json --out out
```

6. Measure the 16 supplied takes. The script finds the raincoat by colour in every take and measures its hue, position, size and the background warmth.

```
python3 prompt_lab.py variance --takes reference/takes --out out
```

7. Read out/variance-by-cell.csv. The variance index should fall from A (1.0) to D. Record which single step (A→B, B→C, C→D) removed the most of each kind of drift.
8. Open out/takes-grid.png and confirm the numbers match what you see row by row.
9. Write out/prompt-notes.md: which fields you will always lock for The Last Ferry, and which you will leave free for variety (for example crowd and steam).

10. Optional extension: generate four takes of your v2 prompt on a fast, cheap draft model at 1K, save them as cellE_01.png ... into reference/takes/ and rerun the variance command. Record the model ID and date.

Verify

v1 exits with status 1 and at least four problems; your v2 exits with status 0; the variance index falls in the order A > B > C > D.

Expected outputs

- out/shot2-fields-v1.prompt.txt and .json with the lint report
- out/shot2-fields-v2.json, .prompt.txt and .prompt.json
- out/takes.csv, out/variance-by-cell.csv, out/takes-grid.png
- out/prompt-notes.md

Evidence checklist

Submit all of the following. Each item is something an assessor can look at.

- ☐ Lint output for v1 and the clean lint output for your v2
- ☐ Your v2 prompt in prose and JSON with @element references
- ☐ Variance table for designs A-D with your reading of it
- ☐ Written list of always-locked and deliberately-free fields

Troubleshooting

Symptom	What to do
'no raincoat found'	The take has no saturated yellow region: check the file is one of the supplied takes or a real generation of MEI.
My v2 still reports a CONFLICT	Remove one of the two terms; describe the camera once.
Lint says REPEATED	Say it once. Repetition dilutes a prompt; it does not emphasise it.
My own generated takes vary more than cell A	Expected: the supplied takes are a designed simulation. Lock the reference images first.

References used in this lab

- [S7] Mira AI, '6 AI Video Tools I'd Use To Make An AI Short Film', YouTube, 14 Jun 2026, youtube.com/watch?v=Wwh8vCqboxQ - *Full video, 10:04*.

A multi-tool workflow: an LLM to plan the story shot by shot, a cinematic image model for character sheets (front, side, back, close-ups) and multi-angle environment grids, a design canvas for the storyboard with keyframe placeholders, a video model for motion, an AI music generator (instrumental, mood and tempo), and an editor; the six-point prompt framework (subject, action, environment, camera, style, constraints); timeline prompting; video extension to pass the previous clip; generations capped at about 15 seconds.

- [S10] Youri van Hofwegen, 'How to Make Your First AI Movie (Full Guide)', YouTube, 11 Jun 2026, youtube.com/watch?v=fs5S867VQzg - *Video description, 17:03*.

Generating the story; building consistent characters and locations; JSON prompting; asset management; video references for connected scenes; editing; fixing voice continuity across clips.

- [S11] ElevenLabs, 'How to Make Your First AI Short Film (Full Tutorial)', YouTube, 9 Aug 2026, youtube.com/watch?v=47DH_VUa67A - *Chapters, 31:13*.

Explore cheaply with faster image models before committing credits; prompt structure across style, subject, objects and negative prompting; a character reference sheet; generate the voiceover FIRST and build the storyboard to match it; direct delivery line by line with audio tags; still frames before video; start and end frames; generate at lower resolution and upscale only the takes you keep; background music; sound effects; edit and trim on a timeline; regenerate with a new character.

- [S13] M. Tschochohei and F. Schenker, 'Multimodal Generative AI in the Enterprise: From Pixels to Profit', Packt, June 2026, ISBN 978-1-80611-167-1 - *Ch.5 pp.56-75; Ch.7 pp.106-119; Ch.8 pp.127-143*.

VAE, GAN and diffusion mechanisms and failure modes; prompt structure and subject lock; binary masks (white = editable), inpainting, outpainting and mask-free editing; spatial versus temporal coherence, 3-D U-Nets and diffusion transformers over spatiotemporal patches; long-running operation polling; responsible-AI risks (voice cloning, deepfakes, likeness) and transparency (C2PA Content Credentials, invisible watermarking, model cards).

- [S15] Google AI for Developers - Gemini API image generation documentation - [*https://ai.google.dev/gemini-api/docs/image-generation](https://ai.google.dev/gemini-api/docs/image-generation) (verified 6 Sep 2026)*.

Native image models for text-to-image and conversational image editing; supported aspect ratios including 16:9, 9:16, 4:5 and 21:9; output sizes up to 4K by model; every generated image carries a SynthID watermark.

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